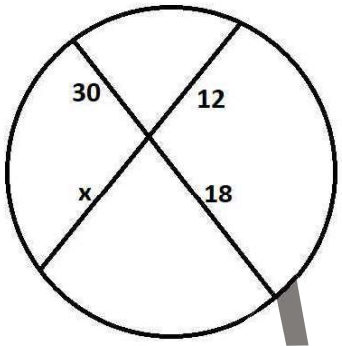
Test #3



1) Find the value of x
   A. 54
   B. 540
   C. 360
   D. 45

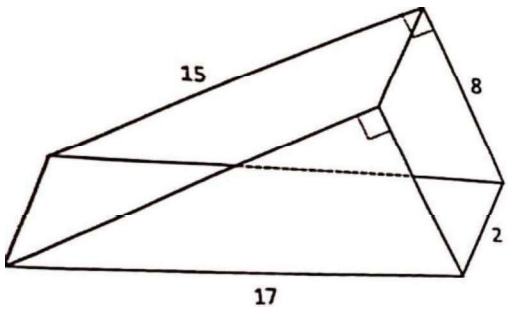
2) Factorize $4x^3y^2(2y^4x + 6x^3y^7)$
   A. $8x^3y^8 + 24x^9y^{14}$
   B. $8x^4y^6 + 24x^6y^9$
   C. $6x^4y^6 + 10x^6y^9$
   D. $6x^3y^8 + 10x^9y^{14}$

3) Minimum points to make a plane.
   A. 4
   B. 3
   C. 2
   D. 1

4) If $-3x + 7 < 16$ which of the following must be true?
   A. $x > -3$
   B. $x < -3$
   C. $x < -9$
   D. $x > -9$

5) What are the solution(s) to the equation $5|x - 2| + 1 = 16$
   A. $x = 3\ only$
   B. $x = 5\ only$
   C. $x = 5\ and\ x = 3$
   D. $x = 5\ and\ x = -1$

6) Find the volume of this shape.
   A. 100
   B. 120
   C. 240
   D. 480

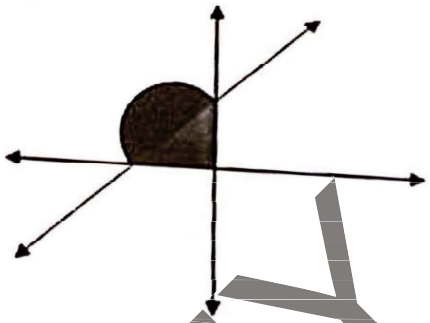


3D ACADEMY

Test #3

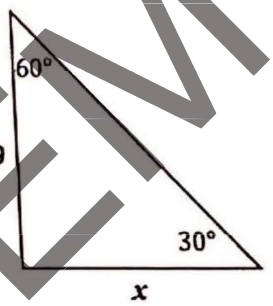
7) Find the perimeter of the following figure. Equation of the line $y = 2x + 4$

A. $\pi\sqrt{5}$
B. 6
C. $6 + \pi\sqrt{5}$
D. $\frac{5}{2}\pi + 4$



8) Find the value of x.

A. $9\sqrt{3}$
B. 4.5
C. 18
D. 9



9) You have a rectangle that its length is 6 more than double its width. The area of this rectangle is 80. Find the value of length.

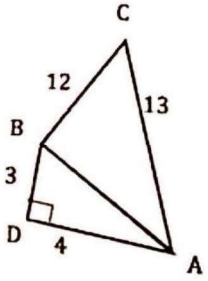
A. 8
B. 16
C. 40
D. 80

10) You have 5 strawberry donuts, 6 chocolate donuts, 3 glaze donuts and 4 caramel donuts. You choose a donut randomly, what's the probability it's strawberry?

A. $\frac{2}{9}$
B. $\frac{1}{6}$
C. $\frac{1}{3}$
D. $\frac{5}{18}$

11) What kind of triangle is the triangle ABC ?

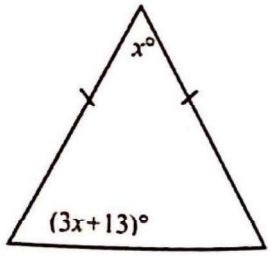
A. Acute triangle
B. Obtuse triangle
C. Right triangle
D. Isosceles triangle



3D ACADEMY

Test #3

12) This is an isosceles triangle. Find the value of x



A. $x = 154$
B. $x = 7$
C. $x = 22$
D. $x = 34$

13) You have a population of rabbits that were first 6 and do they double each year. In 10 years how many rabbits there will be in the population?

A. 12
B. 1200
C. 6144
D. 1024

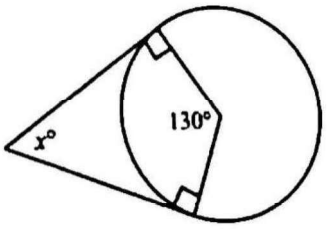
14) Solve the following inequality. $2x - 8 \leq x - 4 < 3(x + 6)$

A. $-11 < x < 4$
B. $-11 > x > 4$
C. $-11 < x \leq 4$
D. $-11 \leq x < 4$

15) If the volume of a cube is equal to 2 times its surface area, what is the value of s?

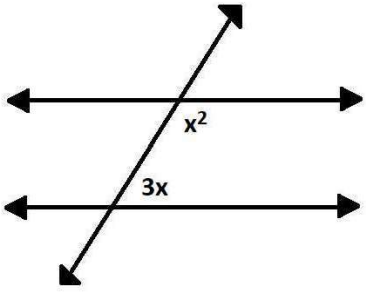
A. $s = -12$
B. $s = 12$
C. $s = -6$
D. $s = 6$

16) find the value of x.



A. $x = 20°$
B. $x = 50°$
C. $x = 90°$
D. $x = 130°$

17) Solve for x



A. 45
B. 180
C. 15
D. 12

3D ACADEMY

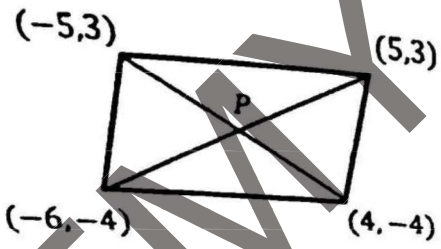
Test #3

18) Solve for x : $2(x+4)=10$

A. 3
B. -1
C. 1
D. 2

19) Find the point in which the two diagonals meet.

A. $(-1,-1)$
B. $(1,1)$
C. $(-\frac{1}{2},-\frac{1}{2})$
D. $(\frac{1}{2},\frac{1}{2})$



20) Which line divides any tringle into two triangles of equal area?

A. A median
B. an altitude
C. an angle bisector
D. A perpendicular bisector of a side

21) In a college with 14,000 students, 490 are majoring in mathematics, what percentage of the student's body dose the number of math major represents?

A. 0.0035%
B. 0.035%
C. 0.35%
D. 3.5%

22) Given that what is $cos^2x=\frac{5}{12}$, what is the value of sin^2x ?

A. $\frac{5}{13}$
B. $\frac{7}{12}$
C. $\frac{5}{7}$
D. $\frac{12}{5}$

3D ACADEMY

Test #3

23) Michael used a compass and a ruler to construct two parallel lines and a transversal. Which of the following statements is a conjecture that Michael can make about the angels formed by the parallel lines and the transversal?

A. Pairs of same side interior angels are supplementary.
B. Pairs of alternate interior angels are supplementary.
C. Pairs of alternate exterior angels are supplementary.
D. Pairs of corresponding angels are supplementary.

24) The endpoints of directed line segment PQ have coordinates of $p(-7,-5)\ and\ Q(5,3)$. What are the coordinates of point A, on $\overline{PQ}$, that divide PQ into a ratio of 1: 3 ?

A. $A(-1,1)$
B. $A(2,1)$
C. $A(3,2)$
D. $A(-4,-3)$

25) Zahra spent $20.50 on 10 party favors for her party. They party favors for each kid was either a puzzle book or a magic trick. The boys each received a puzzle book that cost $1.75 each. The girls each received a magic trick that cost $2.25 each. How mant boys and how many girls attended the party?

A. 4 boys and 6 girls.
B. 5 boys and 5 girls.
C. 6 boys and 4 girls.
D. 7 boys and 3 girls.

26) John is considering accepting one of two sales positions, ABC company offers a yearly salary of $45,000. XYZ Company offers a yearly salary of $38,000 plus 2% annual commission on sales. For what amount of sales (X) in the salary of XYZ company greater than the salary at ABC company?

A. $x > 7{,}000$
B. $x > 35{,}000$
C. $x > 70{,}000$
D. $x > 350{,}000$

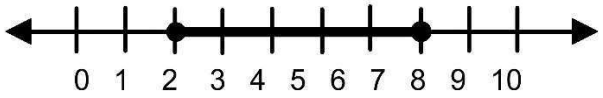
27) Write an equation for the line that has a $y - intercept$ of 2 and is perpendicular to the line $3x + y = 6$

A. $y = -3x + 2$
B. $y = -3x - 2$
C. $y = \frac{1}{3}x + 2$
D. $y = \frac{1}{3}x - 2$

3D ACADEMY

Test #3

28) Which compound inequality is shown by the graph below?



A. $x \geq 2 \; and \; x \leq 8$
B. $x \geq 2 \; or \; x \geq 8$
C. $x \geq 2 \; or \; x \leq 8$
D. $x \leq 2 \; or \; x \leq 8$

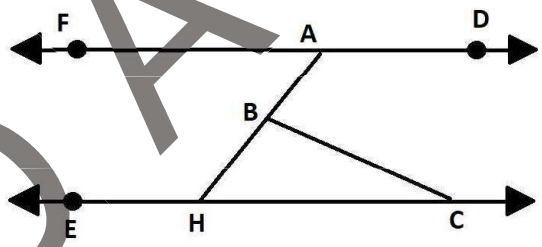
29) Two snow resorts offer private lessons to their customers. Big time sky mountain charges $5 per hour plus $50 insurance. Powder hills charges $10 per hour plus $30 insurance. For what number of hours in the cost of lessons the same for each resort?

A. 3 hours.
B. 4 hours.
C. 5 hours.
D. 6 hours.

30) Find the intersection of the pair of sets.
W = The set of whole numbers
P = The set of nonzero integers

A. $\{\}$
B. $\{0\}$
C. The set of negative integers
D. The set of positive integers

31) In the diagram below $\overline{FAD} \parallel \overline{EHC} \; and \; \overline{ABH} \; and \; \overline{BC} \; are \; drawn$



If $m\angle FAB = 48°$ and $m\angle ECB = 18°$, what is $m\angle ABC$?

A. $18°$
B. $48°$
C. $66°$
D. $114°$

3D ACADEMY

Test #3

32) Employees at the dairy factory are packing cartons of eggs One carton can hold X eggs Today the employees have E eggs to pack. When they have finished thoy have packed C cartons and have 3 eggs left over. Use the equation $\frac{E}{X} = c + \frac{3}{x}$ to find C, the number of the cartons that were packed.

A. $C = \frac{E-3}{x}$

B. $C = \frac{E}{x} - 3$

C. $C = \frac{E}{x-3}$

D. $C = 3 - \frac{E}{x}$

33) In the equation shown below, what is the value of a ?

$$4x^2 + ax = 2x(2x + 1)$$

A. 2

B. 4

C. 6

D. 8

34) If $\angle A$ is an acute angle, how many degrees larger is its supplement than its complement?

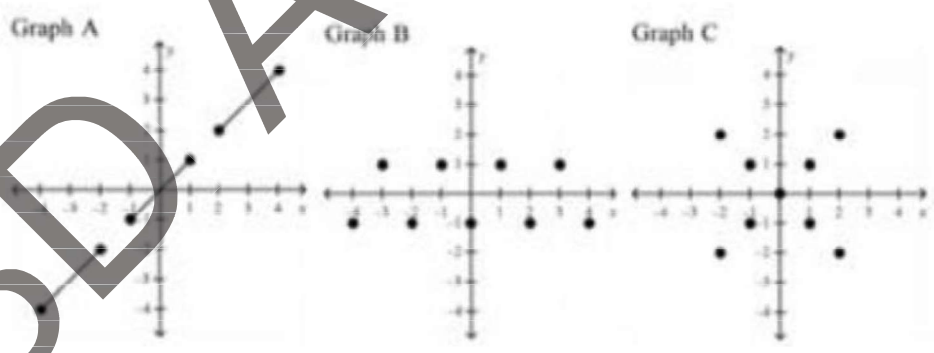
A. $-90°$

B. $0°$

C. $90°$

D. $180°$

35) Determine which of the following graphs represent a function.



A. None of the graphs are functions.

B. All of the graphs are functions.

C. Graphs A and B are functions.

D. Graphs B and C are functions.

3D ACADEMY

Test #3

36) If $0.002 \leq x \leq 0.2$ and $5 \leq y \leq 25$ what is the maximum value of $\frac{x}{y}$?

A. 0.04
B. 0.4
C. 4
D. 40

37) Write an equation in slope-intercept form for the line that passes through (3, 7) and (7, 4).

A. $y = -\frac{3}{4}x + \frac{37}{4}$
B. $y = \frac{3}{4}x + \frac{37}{4}$
C. $y = -\frac{4}{3}x + \frac{37}{4}$
D. $y = -\frac{3}{4}x + \frac{4}{37}$

38) Any true statement has a truth value of 1, and any false statement has a truth value of 0 what is the truth value of this statement?
"All squares are rectangles, or all squares are circles"

A. -1
B. 0
C. $\frac{1}{2}$
D. 1

39) In any right triangle, the altitude to the hypotenuse always :

A. lies inside the triangle
B. lies outside the triangle
C. bisects the hypotenuse
D. bisects the right angle

40) What is the sum of the measures of the interior angles of a 14-sided polygon?

A. 1,980
B. 2,160
C. 2,520
D. 2,880

3D ACADEMY

Test #3

41) Jamie needs to simplify the expression below before she substitutes values for a and b

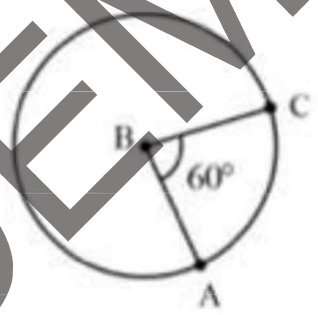
$$\frac{a^{15}b^{12} - a^5b^8}{a^3b^2}$$

If $a \neq 0$ *and* $b \neq 0$which of the following is a simplified version of the expression above?

A. $a^5b^6 - a^5b^4$
B. $a^{12}b^{10} - a^2b^6$
C. a^6b^4
D. a^7b^2

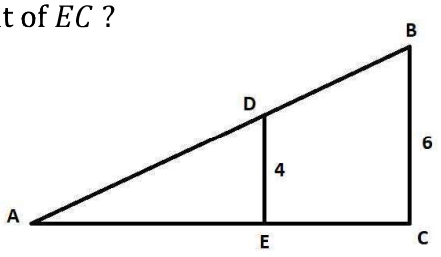
42) If minor $\widehat{AC} = 4\pi\ cm$, what is the area, in square centimeters, of the sector ABC ?

A. 4π
B. 12π
C. $24\,\pi$
D. $40\,\pi$



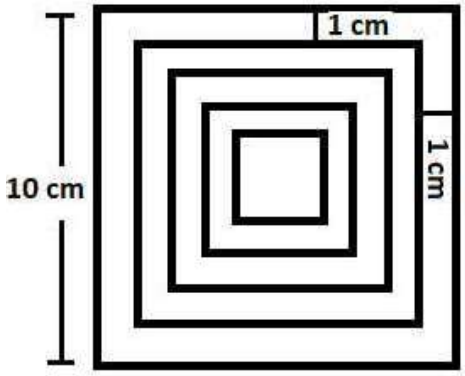
43) In the diagram shown $DE \parallel BC$, DE measures 4 units, BC measures 6 units, and AC measures 9 units. What is the measurement of EC ?

A. 3
B. 5
C. 6
D. $\frac{27}{2}$



44) In the figure, 5 squares are positioned such that the distance between each square is 1 cm, if the side of the largest square is 10 cm, what is the ratio of the area of the largest square to that of the small square?

A. 2:1
B. 5:3
C. 25:9
D. 25:1



3D ACADEMY

Test #3

45) Which expression is equivelent to $(h^2 + 9h - 1)(-4h + 3)$?

A. $-4h^3 - 33h^2 + 31h - 3$
B. $4h^3 + 39h^2 - 23h - 3$
C. $-4h^3 - 39h^2 + 23h + 3$
D. $4h^3 + 33h^2 - 31h + 3$

46) What expression is equivelent to $x^2 - 17x - 60$?

A. $(x - 20)(x + 3)$
B. $(x - 5)(x - 12)$
C. $(x - 20)(x - 3)$
D. $(x + 5)(x - 12)$

47) The formula shown can be used to find A, the amount of money Raed has in his saving account. " $A = p + prt$

Raed wants to find r, the rate of interest his money earns. Which equation is correclty so he find r ?

A. $r = Apt$
B. $r = A - 2pt$
C. $r = \frac{A}{2pt}$
D. $r = \frac{A-p}{pt}$

48) Which expression is equivelent to $\frac{(q^4)^{-3}}{q^{-15}}$ for all values of q where the expression is defined?

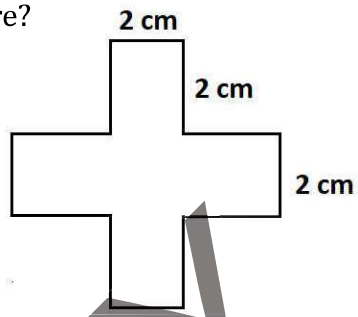
A. q^{27}
B. $q^{\frac{1}{27}}$
C. q^3
D. $\frac{1}{q^3}$

3D ACADEMY

Test #3

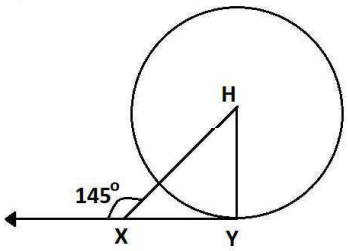
49) The figure is composed of 5 squares. What is the area of the figure?

A. $8\ cm^2$
B. $16\ cm^2$
C. $20\ cm^2$
D. $24\ cm^2$



50) In the circle, H is the center, and $\overline{XY}$ is tangent to the circle, what is the measure of $\angle XHY$?

A. 25°
B. 35°
C. 90°
D. 55°



3D ACADEMY

Test #3

Answer Keys:

Question	Answer	Question	Answer
1	D	26	D
2	B	27	C
3	B	28	A
4	A	29	B
5	D	30	D
6	B	31	C
7	C	32	A
8	A	33	A
9	B	34	C
10	D	35	C
11	C	36	A
12	C	37	A
13	C	38	D
14	C	39	A
15	B	40	B
16	B	41	B
17	D	42	C
18	C	43	A
19	C	44	D
20	A	45	A
21	D	46	A
22	B	47	D
23	A	48	C
24	D	49	C
25	A	50	D

3D ACADEMY